

PCA Implementation

September 19, 2025

1 Principal Component Analysis Implementation

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
import warnings
warnings.filterwarnings('ignore')
```

```
[2]: from sklearn.datasets import load_breast_cancer
cancer = load_breast_cancer()
cancer.keys()
```

```
[2]: dict_keys(['data', 'target', 'frame', 'target_names', 'DESCR', 'feature_names',
'filename', 'data_module'])
```

```
[6]: print(cancer.DESCR)
```

```
.. _breast_cancer_dataset:
```

```
Breast cancer wisconsin (diagnostic) dataset
```

```
-----
```

```
**Data Set Characteristics:**
```

```
:Number of Instances: 569
```

```
:Number of Attributes: 30 numeric, predictive attributes and the class
```

```
:Attribute Information:
```

- radius (mean of distances from center to points on the perimeter)
- texture (standard deviation of gray-scale values)
- perimeter
- area
- smoothness (local variation in radius lengths)
- compactness (perimeter² / area - 1.0)
- concavity (severity of concave portions of the contour)
- concave points (number of concave portions of the contour)

- symmetry
- fractal dimension ("coastline approximation" - 1)

The mean, standard error, and "worst" or largest (mean of the three worst/largest values) of these features were computed for each image, resulting in 30 features. For instance, field 0 is Mean Radius, field 10 is Radius SE, field 20 is Worst Radius.

- class:
 - WDBC-Malignant
 - WDBC-Benign

:Summary Statistics:

	Min	Max
radius (mean):	6.981	28.11
texture (mean):	9.71	39.28
perimeter (mean):	43.79	188.5
area (mean):	143.5	2501.0
smoothness (mean):	0.053	0.163
compactness (mean):	0.019	0.345
concavity (mean):	0.0	0.427
concave points (mean):	0.0	0.201
symmetry (mean):	0.106	0.304
fractal dimension (mean):	0.05	0.097
radius (standard error):	0.112	2.873
texture (standard error):	0.36	4.885
perimeter (standard error):	0.757	21.98
area (standard error):	6.802	542.2
smoothness (standard error):	0.002	0.031
compactness (standard error):	0.002	0.135
concavity (standard error):	0.0	0.396
concave points (standard error):	0.0	0.053
symmetry (standard error):	0.008	0.079
fractal dimension (standard error):	0.001	0.03
radius (worst):	7.93	36.04
texture (worst):	12.02	49.54
perimeter (worst):	50.41	251.2
area (worst):	185.2	4254.0
smoothness (worst):	0.071	0.223
compactness (worst):	0.027	1.058
concavity (worst):	0.0	1.252
concave points (worst):	0.0	0.291
symmetry (worst):	0.156	0.664
fractal dimension (worst):	0.055	0.208

:Missing Attribute Values: None

:Class Distribution: 212 - Malignant, 357 - Benign

:Creator: Dr. William H. Wolberg, W. Nick Street, Olvi L. Mangasarian

:Donor: Nick Street

:Date: November, 1995

This is a copy of UCI ML Breast Cancer Wisconsin (Diagnostic) datasets.
<https://goo.gl/U2Uwz2>

Features are computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. They describe characteristics of the cell nuclei present in the image.

Separating plane described above was obtained using Multisurface Method-Tree (MSM-T) [K. P. Bennett, "Decision Tree Construction Via Linear Programming." Proceedings of the 4th Midwest Artificial Intelligence and Cognitive Science Society, pp. 97-101, 1992], a classification method which uses linear programming to construct a decision tree. Relevant features were selected using an exhaustive search in the space of 1-4 features and 1-3 separating planes.

The actual linear program used to obtain the separating plane in the 3-dimensional space is that described in:
[K. P. Bennett and O. L. Mangasarian: "Robust Linear Programming Discrimination of Two Linearly Inseparable Sets", Optimization Methods and Software 1, 1992, 23-34].

This database is also available through the UW CS ftp server:

```
ftp ftp.cs.wisc.edu
cd math-prog/cpo-dataset/machine-learn/WDBC/
```

.. dropdown:: References

- W.N. Street, W.H. Wolberg and O.L. Mangasarian. Nuclear feature extraction for breast tumor diagnosis. IS&T/SPIE 1993 International Symposium on Electronic Imaging: Science and Technology, volume 1905, pages 861-870, San Jose, CA, 1993.
- O.L. Mangasarian, W.N. Street and W.H. Wolberg. Breast cancer diagnosis and prognosis via linear programming. Operations Research, 43(4), pages 570-577, July-August 1995.
- W.H. Wolberg, W.N. Street, and O.L. Mangasarian. Machine learning techniques

to diagnose breast cancer from fine-needle aspirates. Cancer Letters 77 (1994) 163-171.

```
[7]: df = pd.DataFrame(cancer['data'], columns=cancer['feature_names'] )
```

```
[8]: df.head()
```

```
[8]:
```

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	\
0	17.99	10.38	122.80	1001.0	0.11840	
1	20.57	17.77	132.90	1326.0	0.08474	
2	19.69	21.25	130.00	1203.0	0.10960	
3	11.42	20.38	77.58	386.1	0.14250	
4	20.29	14.34	135.10	1297.0	0.10030	

	mean compactness	mean concavity	mean concave points	mean symmetry	\
0	0.27760	0.3001	0.14710	0.2419	
1	0.07864	0.0869	0.07017	0.1812	
2	0.15990	0.1974	0.12790	0.2069	
3	0.28390	0.2414	0.10520	0.2597	
4	0.13280	0.1980	0.10430	0.1809	

	mean fractal dimension	...	worst radius	worst texture	worst perimeter	\
0	0.07871	...	25.38	17.33	184.60	
1	0.05667	...	24.99	23.41	158.80	
2	0.05999	...	23.57	25.53	152.50	
3	0.09744	...	14.91	26.50	98.87	
4	0.05883	...	22.54	16.67	152.20	

	worst area	worst smoothness	worst compactness	worst concavity	\
0	2019.0	0.1622	0.6656	0.7119	
1	1956.0	0.1238	0.1866	0.2416	
2	1709.0	0.1444	0.4245	0.4504	
3	567.7	0.2098	0.8663	0.6869	
4	1575.0	0.1374	0.2050	0.4000	

	worst concave points	worst symmetry	worst fractal dimension
0	0.2654	0.4601	0.11890
1	0.1860	0.2750	0.08902
2	0.2430	0.3613	0.08758
3	0.2575	0.6638	0.17300
4	0.1625	0.2364	0.07678

[5 rows x 30 columns]

```
[9]: from sklearn.preprocessing import StandardScaler
      scaler = StandardScaler()
      scaler.fit(df)
```

```
[9]: StandardScaler()
```

```
[11]: scaled_data = scaler.transform(df)
      scaled_data
```

```
[11]: array([[ 1.09706398, -2.07333501,  1.26993369, ...,  2.29607613,
                2.75062224,  1.93701461],
              [ 1.82982061, -0.35363241,  1.68595471, ...,  1.0870843 ,
               -0.24388967,  0.28118999],
              [ 1.57988811,  0.45618695,  1.56650313, ...,  1.95500035,
                1.152255  ,  0.20139121],
              ...,
              [ 0.70228425,  2.0455738 ,  0.67267578, ...,  0.41406869,
               -1.10454895, -0.31840916],
              [ 1.83834103,  2.33645719,  1.98252415, ...,  2.28998549,
                1.91908301,  2.21963528],
              [-1.80840125,  1.22179204, -1.81438851, ..., -1.74506282,
               -0.04813821, -0.75120669]])
```

1.1 Applying PCA Algorithms

```
[ ]: from sklearn.decomposition import PCA
      pca = PCA(n_components=2)
```

```
[14]: data_pca = pca.fit_transform(scaled_data)
```

```
[ ]: data_pca    ## after applying the pca we got only 2 columns
```

```
[ ]: array([[ 9.19283683,  1.94858307],
            [ 2.3878018 , -3.76817174],
            [ 5.73389628, -1.0751738 ],
            ...,
            [ 1.25617928, -1.90229671],
            [10.37479406,  1.67201011],
            [-5.4752433 , -0.67063679]])
```

```
[16]: pca.explained_variance_
```

```
[16]: array([13.30499079,  5.7013746 ])
```

```
[17]: plt.figure(figsize=(8,6))
      plt.scatter(data_pca[:,0], data_pca[:,1], c = cancer['target'], cmap='plasma')
      plt.xlabel("First principal component")
      plt.ylabel("Second principal component")
```

```
[17]: Text(0, 0.5, 'Second principal component')
```

